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Predicting the thermal niche of a ubiquitous bacterium using whole genome sequence

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Declaration

I, Aditi Madkaikar, declare that the work done in this project is my own. I have developed the model I present in this work. I gathered the data required for the work from previously published literature, which has been cited appropriately. I received guidance from my supervisor, Prof. Samraat Pawar, and Dr. Arianna Basile over the duration of my project.

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Contents

Abstract	i
1 Introduction	1
2 Methods	5
Selection of model organism	5
Adding enzyme constraints to conventional GEM	5
Temperature dependence of enzyme function	6
Approximation of enzyme function	7
Initial values of temperature dependent parameters	9
Flux Balance Analysis	9
Tuning the model parameters	10
Changing model medium	10
Empirical data for model validation	10
Software and tools used for analyses	11
3 Results	12
Predicting thermal dependence of <i>E. coli</i> growth	12
Parameter sensitivity analysis	13
Model sensitivity to changes in media and reaction bounds	15
4 Discussion	17
References	21
5 Supplementary Information	29
Supplementary Methods	29
Supplementary Results	29

Abstract

Bacteria account for $> 15\%$ of the total global biomass, play a significant role in biogeochemical cycles, and are completely dependent on the environmental temperature for their cellular functioning. There is an increasing need to model the effects of temperature on bacterial metabolism to understand the risk posed by increasing global temperatures to the stability of these cycles. In this study, I propose a method to model the temperature dependence of *Escherichia coli* growth rate using Genome-scale metabolic modelling (GEM). I augmented an *E. coli* GEM (iJO1366) to include enzyme function information for all the enzymes catalysing cellular reactions, and modelled the effects of temperature on these enzymes. The resulting model could predict the thermal performance curve (TPC) of *E. coli* growth rate with high accuracy ($R^2 = 0.94$) over a wide range of temperatures, despite the high level of uncertainty associated with enzyme function data. This model can also account for changes in growth rate due to different cultural media. These results show that this Genome-scale metabolic modelling is a promising method to model microbial TPCs using only genomic information, thus overcoming the challenges faced in quantifying TPCs using traditional culturing techniques.

1 | Introduction

THE phenomenon of climate change has resulted in higher worldwide temperatures and increased frequency and severity of extreme weather events with widespread impacts on natural systems. Temperature plays a vital role in the rates of biochemical and physiological processes of organisms (Fig. 1b). Organisms lacking mechanisms to regulate body temperatures and depending on environmental temperatures to carry out basic life processes like growth and reproduction, such as ectotherms and poikilotherms, are more susceptible to changing temperatures (Deutsch et al., 2008; Huey et al., 2012). To anticipate changes within communities and ecosystems, it's essential to comprehend and predict how they will respond to fluctuations in temperature (Smith et al., 2019).

The changing climate not only influences ecosystems but also has a significant impact on organisms enzymatic activities, and therefore, their metabolic processes. Metabolism provides the energy required for sustenance, growth, reproduction, and other life processes, by harnessing the energy in environmental metabolites, and thus, can be used to determine the demands of an organism on the environment as well as the constraints resource availability places on an organism. Metabolism is regulated such that energy is spent only when necessary (Nelson, 2021). The rate of enzyme function, which is influenced by two key parameters: the optimal temperature (T_{opt}), and melting temperature (T_m), is a limiting factor for cellular metabolism.

Organisms and their cellular metabolism have evolutionarily adapted to their environmental conditions, and exist in a dynamic steady state with their surroundings (Arnold et al., 2001; Clarke, 2003; Prentice et al., 2020; Somero, 1978). Environmental perturbations like thermal fluctuations, pH variation, changes in salinity, etc., impact this state, which in turn affects their metabolism. Phenotypic plasticity (an organism's ability to exhibit different physical or behavioural traits based on external factors), enables an organism to rapidly adapt to changing environments (Huey et al., 2012). Since metabolism is limited by enzyme function, environmental perturbations lead to modification of enzyme function to stabilise the metabolism of the organism. Organisms rapidly modify their enzyme function by changing its activity, in conjunction with the amount of enzyme present in the native state in the cell. This adaptability allows organisms to survive better, thrive in diverse or changing environments, and adjust to short-term temperature variations.

The change in metabolic rate of organisms, in response to change in temperature, can be mapped with a Thermal Performance Curve (TPC), typically using the growth rate as fitness measure (Huey et al., 2012). Typically, TPC is a unimodal response to increasing temperature with peak fitness at the organisms Optimal Growth Temperature (OGT) (Ratkowsky et al.,

1982; Smith et al., 2019; Zwietering et al., 1990). Higher temperatures can denature enzymes and increase the energy required for cellular upkeep (maintenance), and hence are detrimental to a cell's metabolic capabilities (Hobbs et al., 2013; Mateus et al., 2018). Due to the difficulty characterising organism traits over a wide temperature range, a TPC is often modelled using trait proxies (Huey et al., 2012), and while this method is sufficient to capture the TPC's general features, it cannot characterise short-term metabolic thermal responses accurately (Huey et al., 2012; Smith et al., 2019).

Bacteria are prokaryotic poikilotherms occupying $\approx 15\%$ of the global biomass content ($\approx 70Gt$ carbon), and are the second most abundant taxa after plants (Bar-On et al., 2018). They are ubiquitous in terms of the environments they occupy and their ability to metabolise a wide variety of compounds for energy production, survival, and growth (Bar-On et al., 2018; Whitman et al., 1998). As a result, bacteria play a crucial role in global biogeochemical cycles, and nutrient recycling, including soil organic carbon. These cycles are regulated finely by complex interactions occurring within microbial communities (Jansson and Hofmockel, 2020). Thermal fluctuations play a significant role in bacterial metabolism, with increasing temperature generally leading to higher metabolic rates (Brown et al., 2004), and every species responds differently to these fluctuations (Jansson and Hofmockel, 2020). Thus, studying bacterial metabolism is essential to understanding nutrient flow (Bardgett et al., 2008; Ingham et al., 1985; Jansson and Hofmockel, 2020; Paerl and Pinckney, 1996).

A large proportion of environmental bacteria are not culturable in a laboratory environment, and the ones that can be cultured are unable to capture the entire spectrum of bacterial genotypic, and phenotypic variation (Bodor et al., 2020). Over the years, multiple studies have been carried out to study the thermal sensitivity of microbial growth (Bronikowski et al., 2001; Mohr Philip and Krawiec, 1980; Ratkowsky et al., 1982, 2005; Smith et al., 2019), but these studies are limited due to the lack of ability to culture a wide range of bacteria. With the advent of advanced sequencing techniques, whole genomes of previously uncultured bacteria are now freely available (O'Leary et al., 2016; Tatusova et al., 2016). Whole genome availability has led to the use of Machine learning to identify thermal features of bacterial growth (Engqvist, 2018; Gralka et al., 2022a; Li et al., 2019). However, this approach is still unable to characterise the entire TPC of bacterial growth, leading to the need of alternative methods to characterise TPCs based on genomics data.

The knowledge of the whole genome sequence of bacteria has led to the use of a systems-biology based approach to map bacterial metabolic networks. This approach uses the known physical and chemical features of chemical reactions occurring during metabolism and can be graphically represented as a network that maps the links between different reactions (Palsson, 2015; Papin et al., 2003). When the scope of a metabolic network reconstruction reaches a genome level, it is called a Genome-scale metabolic model (GEM) (Fig. 1c). GEMs contain gene-protein reaction (GPR) associations, which act as a way of expressing genotype-phenotype relationships. One

application of GEM reconstructions is to carry out Flux-Balance Analysis (FBA), using optimisation techniques, to predict fluxes through the reactions and to optimise for a value of a particular objective function.

Conventional GEMs do not contain information about anything beyond the stoichiometry of the organism's metabolic network. With the growing advances in the field of reconstructions and other omics-based fields, it is possible to augment the GEMs (Fig. 1d) with a combination of one or more of transcriptomic, proteomic, metabolomic, and fluxomic data as additional constraining layers (Alter et al., 2021; Chen et al., 2020; Grigaitis et al., 2021; Kerkhoven, 2022; Mao et al., 2022; Sánchez et al., 2017). These additions still fall short of GEMs being able to predict the thermal performance curve of bacteria. Recent studies have tried to model the thermal sensitivity of cell growth using GEMs of *Escherichia coli* by associating protein structure with metabolic reaction (Chang et al., 2013) or by modelling protein-folding networks (Chen et al., 2017). Most recently, thermal determinants of *Saccharomyces cerevisiae* (yeast) metabolism were identified using GEMs and thermal experimental data (Li et al., 2021). Li et al. (2021) modelled the thermal performance of yeast growth by describing the thermal sensitivity of enzyme function. This method used a Bayesian learning approach to tackle the uncertainty arising due to the lack of enzymatic data available. Hence, while it is possible to model the thermal performance curve of yeast using GEMs, this method has not been expanded to other organisms.

In this study, I modified and expanded the method proposed by Li et al. (2021), to model the temperature dependence of the growth rate of *Escherichia coli*. The implementation was obtained by integrating two layers (i.e. enzyme function, and its temperature dependence) to constrain an initial Genome-scale metabolic model through an iterative process. Simulations thus carried out allowed me to study the temperature dependence of enzyme function, and its implications on cellular metabolism. It enabled me to identify the most likely effects of climate change on enzyme function, and pave the way for similar studies in the future. Finally, this study aims to present a flexible modelling framework that can enable multi-species interaction extension in the future.

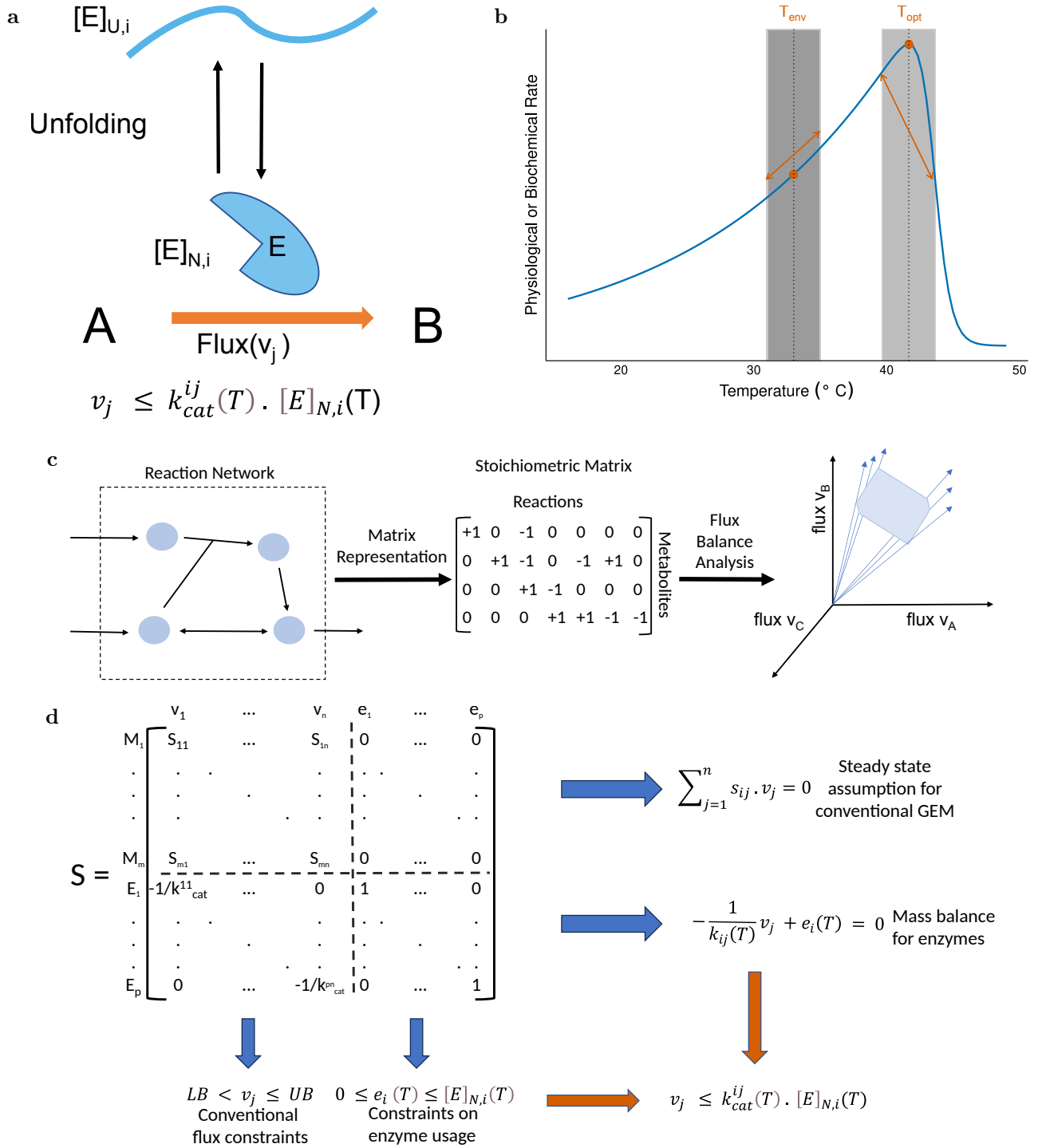


Figure 1: Using Genome Scale Metabolic modelling to predict the temperature dependence of *E. coli* growth

a An illustration of an enzyme-catalysed reaction and the effects of temperature on the reaction flux. The rate of the reaction (v_j) catalysed by the enzyme (E_i) is dependent on the activity of the enzyme (k_{cat}^{ij}), and the amount of enzyme present in the cell ($[E]_{N,i}$). **b** A typical thermal response of organisms traits. Organisms are evolutionarily adapted to the environmental temperature (T_{env}), and perturbations in this cause a change in rate. Typically, the optimal rates are at temperatures (T_{opt}) higher than environmental temperatures. The resulting thermal safety margin protects organisms from death in case of a sudden temperature rise. **c** A network representation of cellular reaction sequence, which in turn can be converted to a mathematical representation in the form of a matrix. This matrix can be used to carry out Flux Balance Analysis (FBA) to optimise the network for a maximum value of growth. **d** Expanding the biochemical stoichiometric matrix to include the enzyme function information, and its temperature dependence.

2 | Methods

IN this study, I incorporated temperature dependence of enzyme function into conventional Genome-Scale Metabolic Model (GEMs) (Li et al., 2021; Sánchez et al., 2017). The organism of choice was *Escherichia coli* (*E. coli*), and iJO1366 was selected as the GEM of interest. First, I integrated the enzyme abundances and activities into iJO1366 to create an ecGEM. The ecGEM was then used as a template to capture the temperature effects on enzymes and carry out Flux Balance Analysis (FBA) simulations to predict growth.

Selection of model organism

In this study, I have used the organism *E. coli* (strain K-12 MG1655) as my model organism. I chose *E. coli*, as it is one of the best-characterised bacteria, and ample empirical data of its Thermal Performance Curve (TPC) is available. It is a versatile organism, and it can grow in a variety of different media under aerobic as well as anaerobic conditions. Due to it being extensively studied, genomic, metabolomic, proteomic, and fluxomic data is available, and its enzymes and their functions are also well-characterised (Palsson, 2015).

Here, I have used a curated conventional GEM (iJO1366) for *E. coli*, which was obtained from the BiGG database (King et al., 2015). It contained 1805 metabolites, 2583 reactions, and 1367 genes (Orth et al., 2011).

Adding enzyme constraints to conventional GEM

I integrated enzyme constraints into iJO1366 using GECKO 3 (Domenzain et al., 2022; Sánchez et al., 2017), to create an enzyme-constrained GEM (ecGEM). GECKO is a tool that can augment enzyme activity and intracellular expression data to a conventional GEM. I chose to add enzyme function data to iJO1366, since I wanted to study the growth rate and cellular metabolism in light of the limitations placed by enzyme activity. The resulting ecGEM operated on the principle that the total flux through an enzyme-catalyzed reaction cannot surpass the enzyme's catalytic capacity, and that the total quantity of enzymes is limited based on empirical data.

Using GECKO I queried the enzyme information from KEGG (Kanehisa and Goto, 2000) and UniProt (Boeckmann et al., 2003), based on the Gene-Protein Relationship (GPR) data present in the conventional GEM. Their activity information (k_{cat}) was queried from BRENDA (Schomburg et al., 2012) using a fuzzy logic matching (see Supplementary Information).

I acquired the proteomic data of *E. coli* containing the intracellular enzyme expression value from the study done by [Mori et al. \(2021\)](#). Additionally, I obtained the absolute protein abundance from PaxDB ([Wang et al., 2015](#)), and it was used to incorporate constraints for enzymes whose expression data was unavailable.

I used GECKO to add the enzymes as pseudo-metabolites, and their usage in a biochemical reaction as pseudo-reactions. These pseudo-reactions were bounded using the intracellular enzyme concentration. In the case of enzymes for which empirical data was unavailable, a pseudo-metabolite representing an aggregated pool of all missing enzymes was introduced. Similarly, a pseudo-reaction drawing mass from this enzyme pool was added Figs. [1c](#) and [1d](#).

After augmenting iJO1366 with enzyme function information, I carried out FBA on the ecGEM to optimise for maximum growth. I also did a Flux Variability Analysis (FVA) to study the variation through all the reaction fluxes (see the section on FBA).

Temperature dependence of enzyme function

The ecGEM modelled the enzyme function using its catalytic activity (k_{cat}), and its total expression in the cell ($[E]_t$). Both of these enzyme properties are temperature dependent. Additionally, the cost associated with cell maintenance (NGAM) increases with increasing temperature ([Li et al., 2021](#); [Smith et al., 2021](#)).

The heat induced denaturation of a protein can be modelled under the assumption that the enzymes in the native and unfolded states are in equilibrium ([2.2](#)), and that the total enzyme content is the sum of the enzyme in native and unfolded states ([2.1](#)).

$$[E]_{t,i} = [E]_{N,i} + [E]_{U,i} \quad (2.1)$$

$$[E]_{N,i} = \frac{1}{1 + e^{\frac{-\Delta G_u(T)}{RT}}} [E]_{t,i} \quad (2.2)$$

Here, $[E]_{N,i}$, $[E]_{U,i}$, and $[E]_{t,i}$ denote the concentration of enzyme in native, unfolded, and total enzyme concentration, respectively, and $\Delta G_u(T)$ denotes the free energy difference between the native and unfolded protein states.

Based on previous studies ([Calhoun et al., 1993](#); [Smith et al., 2021](#)), I modelled the NGAM using an Arrhenius-like function which increased monotonically with temperature, and also accounted for the maintenance cost in case of no growth ([2.3](#)).

$$NGAM(T) = \begin{cases} U_0[1 - B_0 e^{-E_{CUE}/R(\frac{1}{T} - \frac{1}{T_{ref}})}], & \text{if } NGAM(T) \geq 3.2 \\ 3.2, & \text{otherwise} \end{cases} \quad (2.3)$$

Here, U_0, B_0, E_{CUE} denote basal glucose uptake flux, basal carbon use efficiency (CUE), and the activation energy for CUE, respectively, and T_{ref} is the reference temperature. A basal cost of 3.2 for maintenance in case of no growth was used (Calhoun et al., 1993).

The effect of temperature on enzyme activity can be modelled using Eq 2.4, proposed by Hobbs et al. (2013), which is based on the transition state theory (Nelson, 2021).

$$\ln k_{cat}(T) \propto \ln \frac{k_B T}{h} - \frac{\Delta G^\ddagger}{RT} \quad (2.4)$$

Here, $\Delta G^\ddagger$ is the difference in Gibbs free energy between the ground and transition states.

The updated constraints for the FBA to maximise growth were

$$\begin{aligned} Sv &= 0 \\ 0 &\leq v_i \leq k_{cat,i}(T) \cdot [E]_{N,i}(T) \\ \Sigma([E]_{N,i}(T) + [E]_{U,i}(T)) &\leq [E]_t \\ NGAM(T) &= f(T) \end{aligned} \quad (2.5)$$

Approximation of enzyme function

To model the change in enzyme function across a range of temperatures, Equations (2.2) to (2.4) had to be resolved at discrete temperatures. To resolve them, firstly, the change in Gibbs Free energy (ΔG) denoted in terms of change in enthalpy (ΔH) and change in entropy (ΔS) had to be calculated as follows:

$$\Delta G(T) = \Delta H(T) - T\Delta S(T) \quad (2.6)$$

Eq 2.6 was then extended to Eq 2.2, and Eq 2.4 to describe the free energy change associated with protein unfolding, and transition states leading to Eq 2.7, and Eq 2.8, respectively.

$$\Delta G_u(T) = \Delta H^* + \Delta C_{p,u}(T - T_H^*) - T\Delta S^* + T\Delta C_{p,u} \log\left(\frac{T}{T_S^*}\right) \quad (2.7)$$

$$\Delta G^\ddagger(T) = \Delta H_{T_0}^\ddagger + \Delta C_p^\ddagger(T - T_0) + \Delta S_{T_0}^\ddagger + \Delta C_p^\ddagger \ln(T - T_0) \quad (2.8)$$

Here, $\Delta C_{p,u}$ is the heat capacity change between the native and unfolded state, and $\Delta H^*, \Delta S^*$ are the change in enthalpy and entropy at T_H^* & T_S^* , respectively. [Murphy and Gill, 1991](#); [Robertson and Murphy, 1997](#); [Sawle and Ghosh, 2011](#) found the convergence temperatures of ΔH_u & ΔS_u to be $T_H^*(373.5K)$ & $T_S^*(355K)$ respectively. Further, $\Delta H_{T_0}^\ddagger, \Delta S_{T_0}^\ddagger$ & $\Delta C_p^\ddagger$ are the difference in enthalpy, entropy, and heat capacity between the ground and transition states, respectively, and T_0 is the reference temperature.

To fully model the thermal variation in enzyme activity, $\Delta G^\ddagger(T)$, and $\Delta G_u(T)$ need to be resolved, such that they could be approximated using available empirical data. As seen from [Eq 2.7](#) & [Eq 2.8](#), $\Delta H^*, \Delta S^*, \Delta C_{p,u}, \Delta S_{T_0}^\ddagger, \Delta H_{T_0}^\ddagger$, and $\Delta C_p^\ddagger$ need to be estimated to model the protein activity and amount of protein in the native state. These parameters cannot be estimated directly, due to the lack of experimental data, and hence have to be approximated using the thermodynamic properties. They can be approximated using the following six properties:

1. At protein melting temperature(T_m):

$$\Delta G_u(T_m) = 0 \quad (2.9)$$

2. At T_{opt} , enzyme activity is maximized:

$$\left. \frac{dr}{dT} \right|_{T=T_{opt}} = 0 \quad (2.10)$$

$$r = k_{cat}[E]_N$$

3. k_{cat} is known at T_{opt} ([Engqvist, 2018](#)) and:

$$k_{cat}(T_{opt}) = \frac{k_B T}{h} e^{\frac{-\Delta G^\ddagger(T_{opt})}{RT_{opt}}} \quad (2.11)$$

4. ΔC_p can be approximated from the cell growth rate. For *E. coli*, it is approximately $-4kJ/mol/K$ ([Lee and Kaletunc, 2002](#)).

5. Strong linear correlation between ΔH^* & ΔS^* :

$$\Delta H^* = 299.58\Delta S^* + 20008J/mol \quad (2.12)$$

6. For experimentally measured T_{90} (Temperature at which 90% of the enzyme is unfolded):

$$\Delta G_u(T_{90}) = -RT_{90} \ln 9 \quad (2.13)$$

In the absence of experimental T_{90} or $\Delta C_{p,u}$ values, ΔH^* & ΔS^* can be approximated using the length (N) of protein sequence [Eq 2.14](#) ([Sawle and Ghosh, 2011](#)).

$$\begin{aligned} \Delta H^* &= (4.0N + 143) * 1000 \\ \Delta S^* &= 13.27N + 448 \end{aligned} \quad (2.14)$$

Initial values of temperature dependent parameters

The ecGEM included 1366 enzymes, of which the empirical values of T_m for 841 enzymes were available ([Leuenberger et al., 2017](#); [Nikam et al., 2020](#)), and T_{90} for 390 enzymes were available ([Leuenberger et al., 2017](#)). [Leuenberger et al. \(2017\)](#) reported the 95% confidence interval for their findings, and the standard error was calculated to be $4.07^\circ C$. For the remaining enzymes, the T_m was approximated to the mean of the 841 enzymes, which came out to $55.57^\circ C$, and the uncertainty was described as the standard deviation of the empirically obtained T_m s, which was $7.59^\circ C$.

T_{opt} vales for 894 of the 1366 enzymes in the model were available in the dataset by [Li and Engqvist \(2019\)](#). The T_{opt} values for the remaining enzymes were initially set to $37^\circ C$.

The $\Delta C_p^\ddagger$ was approximated using the heat capacity change with cell growth to be -4kJ/mol/K ([Lee and Kaletunc, 2002](#)) for all enzymes in the model. Since this value is an approximation and $\Delta C_p^\ddagger$ values are the property of an enzyme, these values were sampled from a normal distribution with a mean of -4000 and standard deviation of 1000 to ensure the $\Delta C_p^\ddagger$ remain negative.

Flux Balance Analysis

Flux Balance Analysis (FBA) is an application of GEMs that converts the metabolic network into a numeric matrix where the rows represent the metabolites, and the columns represent the reactions. This matrix is stoichiometrically balanced and imposes constraints on the metabolite flow, by accounting for metabolite mass balance. A system of linear equations is created assuming a steady state, and the reaction fluxes are bounded to constrain the solution space thus created. This system is then optimised for a value of a particular objective function ([Price et al., 2004](#)), like the growth rate, Adenosine triphosphate (ATP) production, or any other biologically relevant reaction. Flux Variability Analysis (FVA) is another application of GEMs,

that calculates the variability through each of the reaction fluxes of the model.

For temperatures in the $8 - 48^{\circ}\text{C}$ range, I calculated the NGAM, the enzyme activity, and enzyme quantity, and updated the model accordingly. Then, I carried out an FBA at each temperature point, optimising for maximum growth (see [Supplementary Information](#) for initial simulation results).

Tuning the model parameters

The model had 1367 enzymes, each with three parameters: the temperature of optimal activity (T_{opt}), enzyme melting temperature (T_m), and the change in heat capacity between ground and transition states ($\Delta C_p^{\ddagger}$) amounting to a total of 4101 parameters.

NGAM, enzyme activity (k_{cat}), and enzyme denaturation ($[E]_{N,i}$) were added to the model and varied individually to quantify the significance of each class to the model results. Then each of the enzyme parameters, T_m , T_{opt} , and $\Delta C_p^{\ddagger}$ were varied individually to quantify the effect of each parameter on the model.

Multiple FBA simulations were carried out at each step, and for each simulation, the predictions were compared with *E. coli* TPC obtained from the TPC database gathered by [Smith et al. \(2019\)](#). I then calculated the R^2 and Mean Square Error (MSE) scores for the model and the empirical data were calculated for each simulation to quantify the goodness of each prediction, and update the parameter (T_m , T_{opt} , and $\Delta C_p^{\ddagger}$) estimation. The complete algorithm used in parameter tuning is described in [Supplementary Information](#).

Changing model medium

I modified the growth medium of the ecGEM and the bounds on reactions denoting intracellular enzyme availability, and ran multiple FBA simulations to check the relative effect of nutrient availability, and enzyme availability on growth rate. First, I modified the medium to a nutrient-poor minimal medium (MM), then changed it to a nutrient-rich Lysogeny broth (LB) medium. I ran the following four FBA simulations for each media: Normal Media Normal Bounds, Normal Media Changed Bounds, Changed Media Normal Bounds, and Changed Media Changed Bounds. I obtained the LB media composition from [Machado et al. \(2018\)](#), and the MM composition using COBRApy ([Ebrahim et al., 2013](#)).

Empirical data for model validation

To validate the model predictions, I used the TPC database of bacteria curated by [Smith et al. \(2019\)](#), containing 604 unique TPCs. I filtered the dataset for *E. coli*, and my validation dataset contained 26 unique TPCs for *E. coli* growth. Among the 26 unique TPCs, I chose the one

previously reported by [Mohr Philip and Krawiec \(1980\)](#) to train my model, as it contained *E. coli* growth rate data over a wide temperature range ($20 - 48^{\circ}\text{C}$), had reported a growth medium, and also quantified the steep decline in growth at very high temperatures. I used the remaining 25 TPCs to validate my model predictions.

Software and tools used for analyses

Since GECKO was implemented in MATLAB ([The MathWorks Inc, 2022b](#)), I used MATLAB (v9.13.0.2126072) to create the ecGEM. I carried out the temperature parameter modelling using Python (v3.10.0) ([Van Rossum and Drake, 2009](#)), and modified the ecGEM using COBRApy (v0.26.3) ([Ebrahim et al., 2013](#)). All the FBA were carried out using Gurobi LP solver (v9.5.1rc2) ([Gurobi Optimization, LLC, 2023](#)).

The statistical tests for model performance were carried out using Scikit-learn ([Pedregosa et al., 2011](#)). The model prediction visualisations were done using tidyverse ([Wickham et al., 2019](#)), ggplot2 ([Wickham, 2016](#)), and cowplot ([Wilke, 2020](#)) in R (v4.1.2) ([R Core Team, 2021](#)).

3 | Results

IN this study, I developed and curated a method to predict how temperature changes impact the growth rate of *Escherichia coli*. I created a model that considers the impact of temperature on enzyme activity and denaturation, influencing cellular metabolism. Using iJO1366 (King et al., 2015; Orth et al., 2011) as a template, I developed an enzyme-constrained Genome-scale metabolic model (ecGEM) for *E. coli*. Initially, temperature-dependent simulations didn't accurately predict growth rates across temperatures (Figs. S2a to S2c). Adjusting enzyme-specific temperature parameters, however, led to improved accuracy ($R^2 = 0.94$; Fig. 2c). The model performed well over a wide range of temperatures ($10 - 48^\circ\text{C}$) (Fig. 2), accurately accounting for alterations of the growth rate caused by changes in growth media (Fig. 4).

Predicting thermal dependence of *E. coli* growth

I carried Flux Balance Analysis (FBA) on the fully trained temperature variant enzyme constrained model (ecGEM), and it predicted a maximum growth rate of 1.49 at 38.5°C ($R^2 = 0.94$, and $MSE = 0.009$; Fig. 2a), whereas the empirical data showed a maximum growth rate of $1.44h^{-1}$ at 38.5°C (Fig. 2a). This final model accurately predicted the increasing growth with temperature, the maximum growth rate, the temperature of maximum growth rate, and the steep decline in growth at higher temperatures. It continued to show good predictions in the low-temperature range ($< 20^\circ\text{C}$) (Fig. 2c), and was also able to predict the change in CO_2 , O_2 , glucose, and acetate fluxes with temperature (Fig. 2b). However, it was unable to accurately predict the maximum growth rate achieved in a validation TPC showing a high empirical growth rate ($2.4h^{-1}$) (Fig. 2d).

The empirical data showed that the temperature of maximum growth rate is 38.5°C , and beyond 46.5°C , there is a steep decline in growth rate. The final values for enzyme T_{opt} show a slightly right-skewed distribution (Fig. 3c), having a range of $28.5 - 42.3^\circ\text{C}$, with mean of 42.2°C . Similarly, values for enzyme T_m follow a right-skewed distribution (Fig. 3c), having a range of $40.8 - 85.18^\circ\text{C}$, with a mean of 55.7°C . The final parameter values show that T_m of 70 enzymes is lower than 45°C , whereas the initial parameters showed 46 enzymes. Initial parameter values showed T_{opt} of 1262 enzymes is lower than 38.5°C , while the final parameter set had 148 enzyme with $T_{opt} < 38.5^\circ\text{C}$.

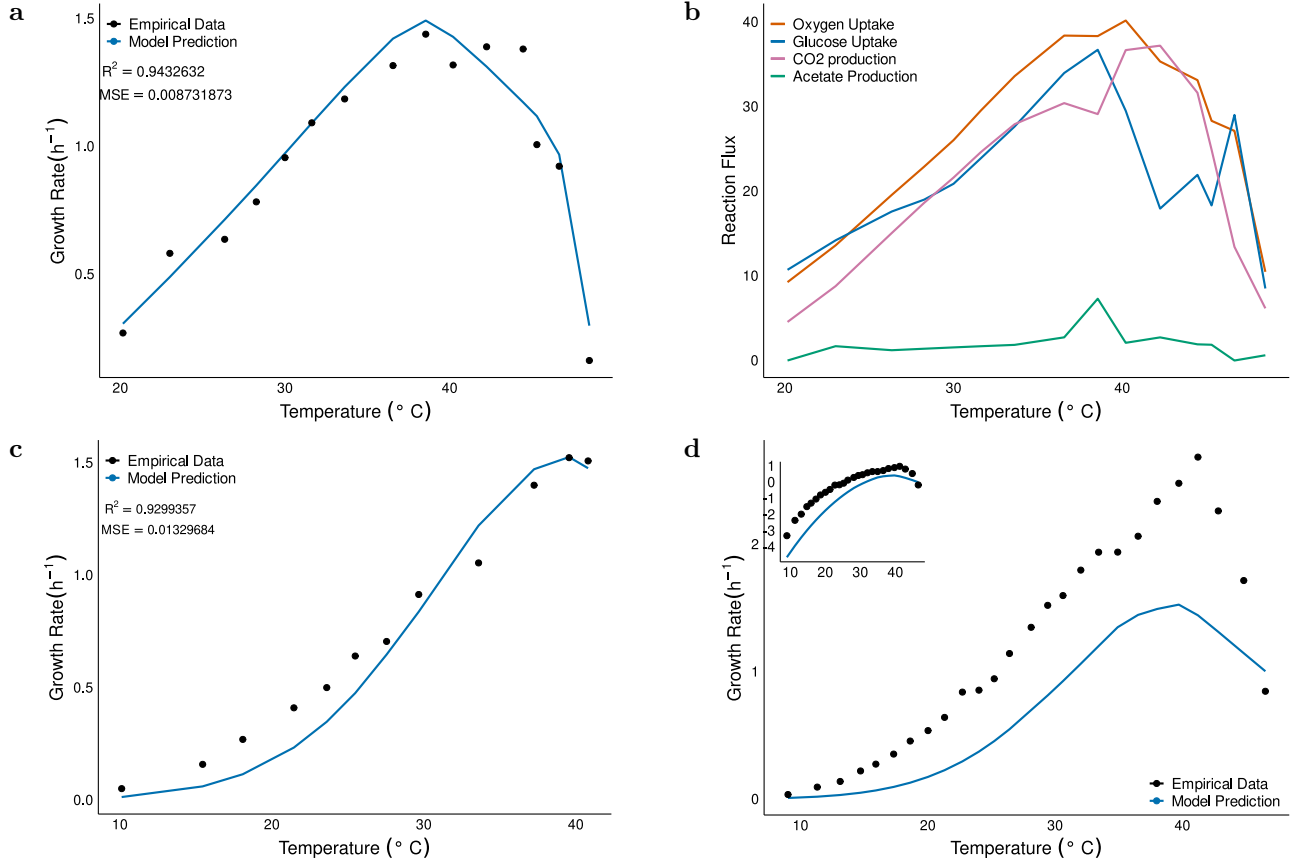


Figure 2: **Model results with final parameter values**

a Flux Balance Analysis (FBA) run with final parameter values, and the predictions compared with the data from the training dataset. The parameter values showed $R^2 = 0.94$, and $MSE = 0.009$. **b** The predicted fluxes for glucose, O_2 uptake, and CO_2 , and acetate production using the final parameter values. **c** Using the model to predict the growth rate using a validation dataset including the low-temperature range ($10 - 20^\circ C$). **d** Using the same model to predict the growth on a validation dataset having a with maximum growth rate value. Inset shows the same validation dataset's $\log(\text{growth})$ vs temperature.

Parameter sensitivity analysis

The temperature variant ecGEM contained three factors that change with temperature, namely: non-growth associated maintenance cost ($NGAM(T)$), enzyme activity ($k_{cat}(T)$), and enzyme denaturation ($[E]_{N,i}(T)$). Effects of these factors were observed by running multiple simulations by adding individual factors to the ecGEM (Fig. 3a). Variations in the $NGAM(T)$, affect the growth at higher temperatures only, whereas the addition of $k_{cat}(T)$ resulted in the unimodal shape of the prediction curve (Fig. 3a). $[E]_{N,i}(T)$ led to a drastic fall in growth rates across a wide range of temperatures (Fig. 3a).

The initial parameter values for the 1367 enzymes present in the ecGEM were estimated as described in Methods. Each enzyme was described using three key parameters: temperature of optimal activity (T_{opt}), melting temperature (T_m), and difference in heat capacity between ground and transition states ($\Delta C_p^\ddagger$), led to a total of 4101 temperature variant parameters.

The empirically measured values of T_m followed a normal distribution with a mean of 55.6°C (Fig. S2d), and T_{opt} values also showed a normal distribution with a mean of 35.4°C (Fig. S2d). The effects of each enzyme parameter (T_{opt} , T_m , and $\Delta C_p^\ddagger$) on the model were observed by modifying them and running multiple FBA. Variations in T_{opt} (see Methods) led to a change in the temperature of maximum growth, whereas changes in T_m (see Methods) led to a drastic drop in growth rates in a few simulations (Fig. 3b). Changes in $\Delta C_p^\ddagger$ did not lead to a significant change in model predictions (Fig. 3b). Finally, relaxing the bounds on the pseudo-reaction for enzyme usage, led to a rise in growth rate over the entire temperature range (Fig. S2c).

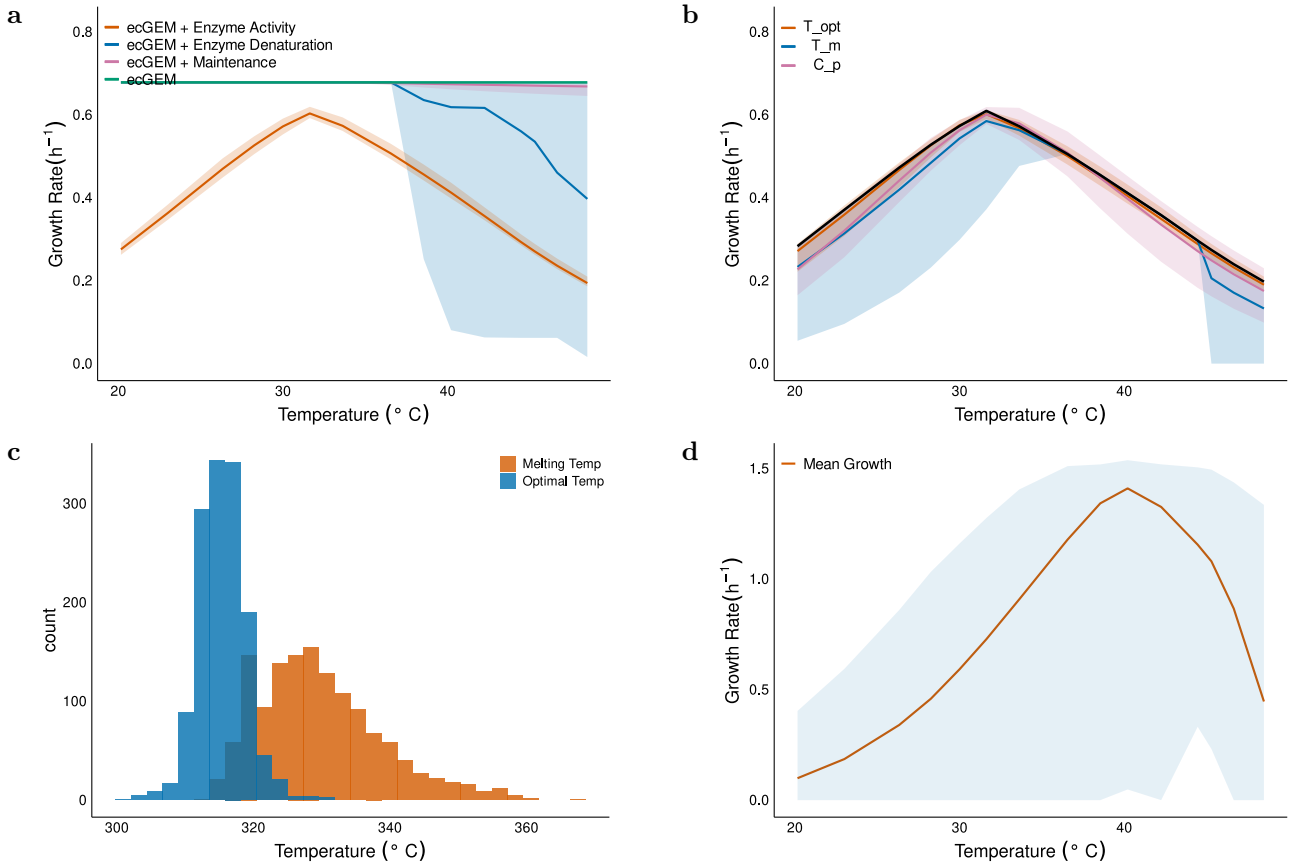


Figure 3: Parameter Sensitivity Analysis

a Effect of each temperature dependent factor on the ecGEM. Fig. legend: ecGEM- predictions over the temperature range using the enzyme constrained GEM, ecGEM + Enzyme Activity- Addition of effects of temperature dependence of enzyme activity to ecGEM, ecGEM + Enzyme denaturation- Addition of effects of enzyme denaturation to ecGEM, ecGEM + Maintenance- Addition of effects of thermal dependence of cell maintenance to ecGEM **b** Effect of sampling T_{opt} , T_m , and $\Delta C_p^\ddagger$ on the temperature variant ecGEM. Fig. legend: T_{opt} - effect of sampling values of optimal enzyme activity for all enzymes from a distribution $N(T_{opt}, 4)$, T_m - effect of sampling values of enzyme melting temperature from a distribution $N(T_m, T_{mstd})$, $\Delta C_p^\ddagger$ - the effect of sampling the difference in heat capacity between ground and transition state from a distribution $N(-4000, 1000)$. **c** The distribution of T_{opt} , and T_m in the final parameter estimates. The mean of the T_{opt} was 42.3°C , and that of T_m was 55.7°C . **d** The range of growth rate achieved by sampling the values for T_{opt} , T_m , and $\Delta C_p^\ddagger$. The shaded region represents the solution space created by sampling, and the line represents the mean growth rate.

Model sensitivity to changes in media and reaction bounds

I used the final parameter values to model the changes in growth rate due to a change in the growth medium. Changing the media to a nutrient rich media, Lysogeny broth (LB), led to a slight increase in model prediction accuracy (Original prediction $R^4 = 0.45$, $MSE = 0.36$; Prediction with media change $R^2 = 0.5$, $MSE = 0.33$) (Fig. 4a). On the other hand, relaxing the bounds placed on the reactions denoting intracellular enzyme availability, led to significantly improved predictions ($R^2 = 0.83$, $MSE = 0.11$) (Fig. 4a). Combining the effects of media change with bound relaxation resulted in the best predictions ($R^2 = 0.88$, $MSE = 0.08$) when validation was performed with a TPC showing high maximum growth (Figs. 4a and 4c). Changing the media to a nutrient poor media, Minimal Media (MM), led to a sharp decrease in growth rate prediction over the entire temperature range (Fig. 4d). For the minimal media, significant improvement was seen upon the change in media (Original prediction $R^4 = -1263.4$, $MSE = 1.26$; Prediction with media change $R^2 = 0.28$, $MSE = 0.00072$) (Fig. 4b). Model prediction with media change coupled with constraining the bounds placed on the reactions denoting intracellular enzyme availability ($R^2 = 0.22$, $MSE = 0.00077$), did not show a significant improvement to the model predictions with only media change (Fig. 4b). Similarly, only constraining the reaction bounds did not improve predictions (Fig. 4b).

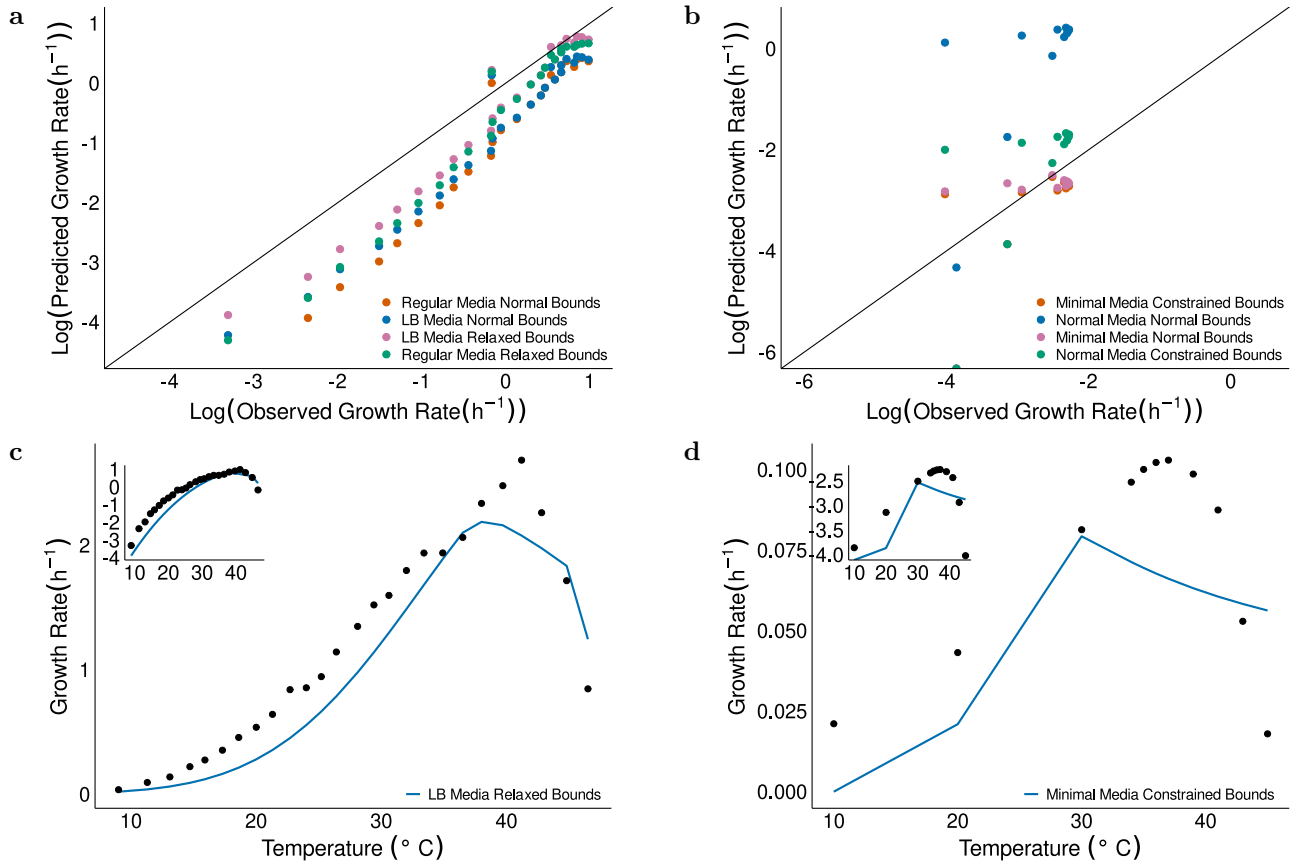


Figure 4: Model sensitivity and reaction bounds changes

a Effect of changing growth media to Lysogeny broth (LB) and changing bounds of reactions denoting intracellular enzyme availability. Fig. legend: Regular Media Normal bounds- No changes to trained temperature dependent model, LB Media Normal bounds- Effect of changing media to Lysogeny broth (LB), LB Media Relaxed bounds- Effect of changing media to Lysogeny broth (LB) and relaxing the bounds on the reactions denoting intracellular enzyme availability, Normal Media Relaxed bounds- Effect of relaxing the bounds on the reactions denoting intracellular enzyme availability **b** Effect of changing growth media to Minimal media (MM) and changing bounds of reactions denoting intracellular enzyme availability. Fig. legend: Minimal Media Constrained bounds- Effect of changing the media to MM and constraining the bounds placed on reactions denoting intracellular enzyme availability, Normal Media Normal bounds- No changes to trained temperature dependent model, Minimal Media Normal bounds- Effect of changing media to MM, Normal Media Constrained bounds- Effect of constraining the bounds on the reactions denoting intracellular enzyme availability **c** Flux Balance Analysis (FBA) run with relaxed bounds, and LB, and predictions compared with validation dataset having high maximum growth. Inset shows the same graph in log scale. **d** Flux Balance Analysis (FBA) run with constrained bounds, and MM, and predictions compared with validation dataset having low overall growth. Inset shows the same graph in log scale.

4 | Discussion

IN this study, I present a method to model and predict the effects of temperature changes on the cell growth rate (TPC) of *Escherichia coli*. I modelled the effects of temperature on enzyme activity and denaturation, which in turn affect the rate of cellular metabolism. Using iJO1366 (King et al., 2015; Orth et al., 2011) as a template for the *E. coli* metabolism network, I created an enzyme-constrained genome-scale metabolic model (ecGEM), which I used to model temperature effects on individual enzyme functions. The initial temperature-dependent simulations could not accurately predict the growth rate of *E. coli* over a wide range of temperatures (Figs. S2a to S2c). It showed poor results in predicting the maximum growth rate, as well as the temperature at which it occurs. However, after modifying the values of the enzyme specific temperature dependent parameters, the resulting model could accurately predict the growth rates across a wide range of temperatures (Fig. 2c). The model was also able to predict the TPCs of *E. coli*, in different growth media (Figs. 2b, 4c and 4d). Along with the growth rates, this approach could also predict the carbon dioxide (CO_2) and acetate production fluxes (Fig. 2b), and glucose and oxygen (O_2) uptake fluxes. It shows that despite the lack of data and the uncertainty associated with empirical proteomic data, it is possible to predict temperature induced changes in bacterial metabolism with high accuracy.

The addition of enzyme activity, and intracellular availability information to iJO1366, placed additional bounds on the flux space, thus over-constraining the resulting ecGEM. Modelling the function of all enzymes across a wide temperature range had a high level of uncertainty associated with it due to the approximations used (see Methods). As a result, the initial temperature variant model performed poorly. Modifying the three temperature-dependent terms, non-growth associated maintenance ($NGAM(T)$), enzyme activity ($k_{cat}(T)$), and enzyme denaturation ($[E]_{N,i}(T)$) individually, led to insights on their effects on ecGEM. The addition of $NGAM(T)$ to ecGEM affected the growth at higher temperatures but had a minor effect at low temperatures (Fig. 3a), suggesting that the cost of maintaining cellular function has a more significant impact at high temperatures. Addition of temperature dependence of enzyme activity (k_{cat}) to the ecGEM led to a unimodal response to temperature (Fig. 3a), indicating that the enzyme activity plays a big role in the rate of cellular metabolism over the entire temperature range, whereas the addition of enzyme denaturation ($[E]_{N,i}$) to the ecGEM also led to a decrease in growth rate at higher temperatures but had minor effects on growth at lower temperatures (Fig. 3a), suggesting that while the enzyme denaturation plays a role on cellular metabolism its effects are apparent only at higher temperatures.

Since the initial parameter estimates showed a low confidence level, individual enzyme

parameters: melting temperature (T_m), optimal temperature (T_{opt}), and difference in heat capacity between ground and transition states ($\Delta C_p^\ddagger$) were modified to study their effects on the growth rate. There was no correlation between T_m , and T_{opt} (Fig. S3), suggesting that they could be varied independently. Varying the $\Delta C_p^\ddagger$ did not have a big effect on growth (Fig. 3b). On the other hand, varying the T_{opt} , and T_m had bigger impacts on the model prediction (Fig. 3b). Changes in T_{opt} led to changes in the temperature of maximum whereas changes in T_m were responsible for a sudden drop in growth. Using these observations to update the parameter values led to improved model predictions (Figs. S2a and 2a).

The final estimation of the enzyme parameters showed that the mean value of T_m was $55.7^\circ C$, and that of T_{opt} was $42.3^\circ C$. A recent study has shown that the average melting temperature (T_m) of *E. coli* enzymes was $55.6^\circ C$ (Mori et al., 2021), showing that the parameter estimation was in agreement with empirical data. The TPC of growth rate data obtained from Smith et al. (2019), showed maximum growth ($1.4h^{-1}$) occurring at $38.5^\circ C$ (OGT), and a steep decline in growth rate beyond $46^\circ C$. The mean T_{opt} of enzyme activity was close to the OGT, as well as the temperature of declining growth. At the same time, the average T_m of enzymes was well above the temperature at which the decline in growth occurs. This suggests that while the T_{opt} of enzyme activity is likely the major contributing factor to the temperature of maximum growth, a combined effect of T_m , and T_{opt} is the cause of the decline in growth at very high temperatures.

Media sensitivity analysis showed that the change in growth medium to the nutrient-poor minimal medium (MM) resulted in a steep decline in growth rate prediction over the entire temperature range (Fig. 4d). The results align with the expectation that changing to minimal media will cause a decline in the overall growth rate. On the other hand, changing the medium to a nutrient-rich Lysogeny broth (LB) medium led to only a slight increase in growth rate (Fig. 4a). This indicated that in nutrient-poor environments, growth is limited by nutrient availability, not enzyme availability (Fig. 4b). However, an abundance of nutrients doesn't result in high growth rates, and the intracellular enzyme availability limits the maximum growth in case of a nutrient-abundant environment (Fig. 4c).

While the modelling approach presented here highlights a promising method to study the TPC of *E. coli*, it has a few drawbacks. It requires protein expression data, which is not readily available and has a high level of uncertainty. Also, the use of macromolecular rate theory (Hobbs et al., 2013), and enzyme denaturation to describe the variation in enzyme function may not be the best approach, as it may be too simplistic to be applied to all enzymes, and has not been extensively validated. Since the growth rate dropped drastically at very high temperatures (Figs. S2a and 2a), it created a large error in initial model predictions, and the model training across the temperature range could be biased to account for this error. The modelling technique can be further improved by using empirical data for cell maintenance cost, extensive proteome profiling over a range of temperatures, carbon uptake and emission fluxes,

as well as by improving the enzyme temperature dependence formulation.

At its core, this approach requires genomic data for creating the metabolic model (Palsson, 2015). While the addition of enzyme function information is an essential step, it does not require empirical enzyme expression data (Sánchez et al., 2017) and can be carried out using enzyme abundance data. Finally, it requires empirical TPC to improve the parameter estimations and validation. Overall, the approach presented here has the potential to be extended to predict the thermal performance of other species of bacteria like *Bacillus subtilis*, *Shigella flexneri*, *Lactococcus lactis*, among others, which have well-characterised genomes, and proteomes.

Bacteria and their metabolic network have evolved to function optimally at temperatures slightly higher than their environment (Arnold et al., 2001; Engqvist, 2018; Somero, 1978). This gives the bacteria a thermal safety margin, which is an important aspect of their response to changing temperatures (Smith et al., 2019) as it protects them from death in case of a sudden increase in temperature (Fig. 1b). Increasing global temperatures have raised concerns over its effects on bacterial metabolism, as they are expected to have higher metabolic rates, and thus impact nutrient recycling, and CO₂ emissions (Smith et al., 2019, 2021). Studying and predicting the microbial community response to fluctuating temperatures is an important aspect of understanding and tackling the problem of climate change (Jansson and Hofmockel, 2020; Smith et al., 2019). Resolving inter-species interactions like competition, cooperation, mutualism, and even horizontal gene transfer, in a microbial community is a persistent ecological problem due to the complexity of microbial behaviour. Conventional methods fall short of predicting such microbial community responses to climate change, and hence, alternate methods to study these communities are being explored (Bodor et al., 2020). Genome-scale metabolic modelling is one method with a high potential to characterise bacterial TPCs as it requires genomic, and not bacterial culture data. As seen from this study, and previous studies (Chang et al., 2013; Chen et al., 2017; Li et al., 2021), it can be used to characterise the thermosensitivity of the growth of individual organisms, but its applications, to date, have been limited to single species. In the future, this method could be used to model microbial communities by incorporating the thermal response of each species in the community.

In conclusion, it can be seen that Genome-scale metabolic models are a viable method to study the effects of temperature on bacterial growth, and metabolism. This approach not only provides a link between physiological, and biochemical traits of an organism, but also can be implemented with limited data. This study can be viewed as a pilot study in the direction of using GEMs to predict the effects of changing temperatures on whole microbial communities.

Data and Code Availability

The code, and data required for this analysis are available on [GitHub](#).

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5 | Supplementary Information

Supplementary Methods

Fuzzy logic matching for querying enzyme information

Enzyme activity is organism, and substrate specific. Due to the difficulty in characterising it accurately, a fuzzy logic matching algorithm was used by GECKO to query enzyme activity information, and it was:

1. correct organism, correct substrate
2. any organism, correct substrate
3. correct organism, any substrate
4. any organism, any substrate

Algorithm for modifying T_{opt} , and T_m

Due to the vast parameter space created by the 4101 temperature variant parameters used to describe enzyme function, modifying them systematically became necessary. I used the following algorithm to modify them:

1. Relaxing the bounds placed on enzyme pseudo-reactions
2. Increasing T_{opt} values for all enzymes m times till maximum possible R^2 is reached
3. Sampling T_m n times from a distribution to improve R^2
4. Sampling T_{opt} n times from distribution to further improve R^2
5. Repeating steps 3 & 4 k times to get the best possible prediction

For this study, I carried out multiple runs with $m = 5$, $n = 10$, and $k = 2$; $m = 10$, $n = 100$, and $k = 2$, and $m = 10$, $n = 50$, and $k = 5$. Best results were obtained with the last set of simulations.

Supplementary Results

Running a flux balance analysis with initial parameters

I incorporated the enzyme function information of 1366 enzymes into iJO1366 to create an enzyme-constrained GEM (ecGEM). The resulting ecGEM had 6810 reactions and 3172

metabolites Table S1. This ecGEM model was tuned to account for inaccurate values of enzyme activity (k_{cat}) obtained due to the querying algorithm. The k_{cat} of 14 enzymes was modified Table S2 for the ecGEM to reach the required growth of $0.68h^{-1}$ (Mori et al., 2021). The FVA showed that the variability in the reaction fluxes of ecGEM reduced significantly Fig. S1, and thus ecGEM was a highly constrained metabolic model of *E. coli* containing enzyme function information. The enzyme function in this ecGEM was modified (as shown in methods) to predict growth across a wide range of temperatures ($20 - 48^{\circ}C$). The resulting thermal performance curve (TPC) was unimodal, showing a maximum growth rate of $0.6h^{-1}$ at $\approx 32^{\circ}C$ ($R^2 = -1.8$, and $MSE = 0.44$ Fig. S2a). The empirical data showed maximum growth rate of $1.4h^{-1}$ at $38.5^{\circ}C$ (Fig. S2a). The model showed good predictions for temperatures lower than $25^{\circ}C$ (Fig. S2b), capturing the unimodality of a TPC. However, it could not predict the maximum growth rate, the temperature at which maximum growth occurs, and the steep decline in growth at very high temperatures ($> 46^{\circ}C$). It was also unable to predict the growth of *E. coli* in different media (Fig. S2c).

	iJO1366	ecGEM
General Description:		
Number of reactions	2583	6810
Number of metabolites	1805	3172
Number of genes	1367	1367
Number of compartments	3	3
Reactions:		
Exchange reactions	324	324
Transport reactions	311	2215
Enzyme usage reactions	0	1366
Isozyme catalysed reactions	0	1970
Metabolites:		
Original	1805	1805
Enzymes	0	1367
Standard enzyme	0	1

Table S1: Comparison between iJO1366 and ecGEM

Protein	Modified k_{cat}
P31119	120
P0A7E3	56
P0A6A8 + P31119	25
P0A9D4	3.4
P0ADF6	236.2775
P17854	1.675
P17854 + P0AC59	1.675
P17854	1.675
P02932	15000
P0A7B1	235
P0AEK2	29.5
P0AGB0	75
P0AAI5	1.175
P25516	132.5

Table S2: Proteins and their modified k_{cat} values

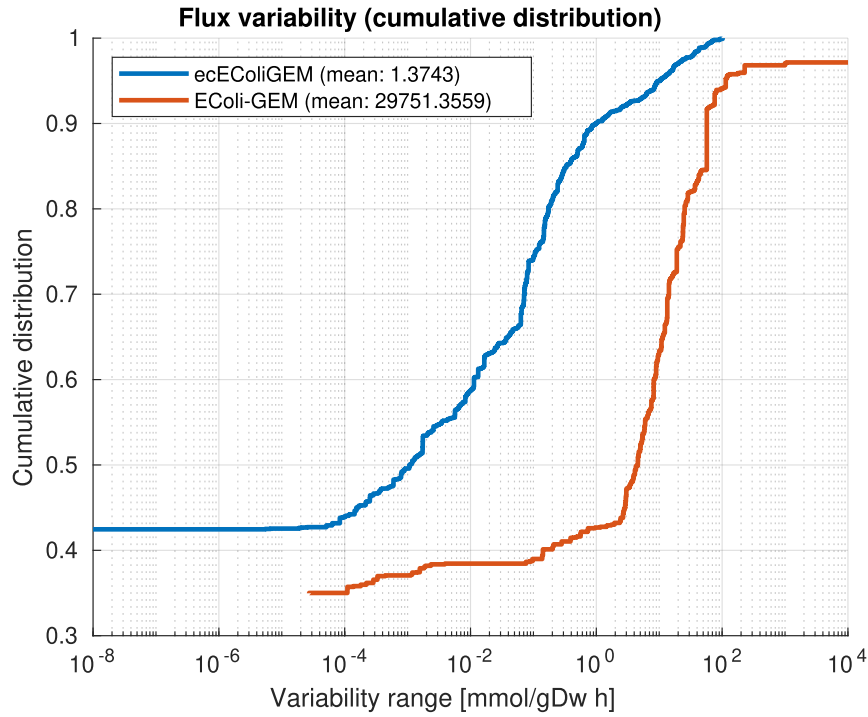


Figure S1: ecGEM reducing flux variance

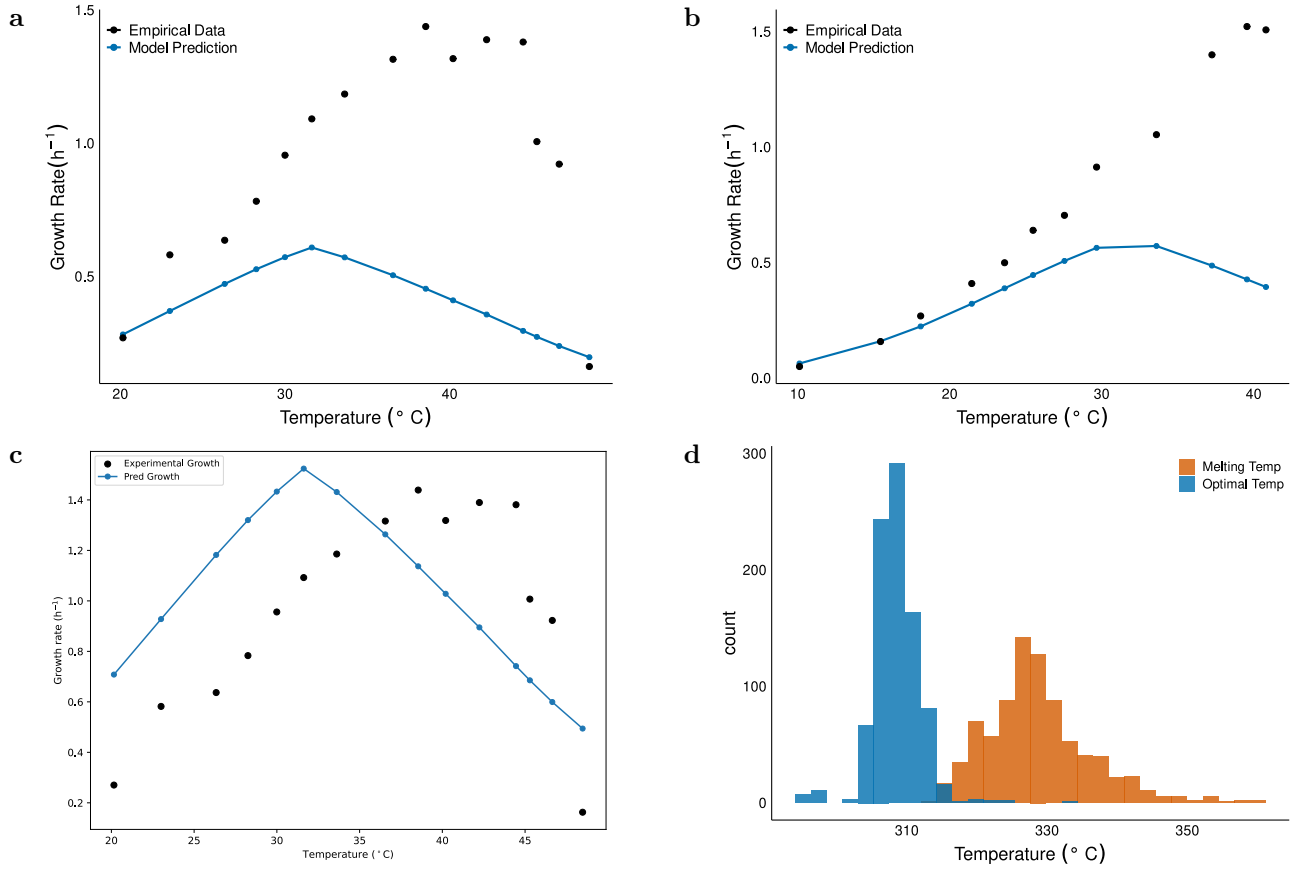


Figure S2: **Flux Balance analysis (FBA) simulation at discrete temperatures using initial parameters**

a Flux Balance Analysis (FBA) between 20 – 48 $^{\circ}\text{C}$, and the predictions compared with the data from the training dataset. **b** Using the same initial parameters to predict the growth rate using a validation dataset including the low-temperature range (10 – 20 $^{\circ}\text{C}$). **c** Initial parameter values for empirically measured T_{opt} , and T_m . **d** Effect of relaxing bounds on reactions denoting intracellular enzyme abundance.

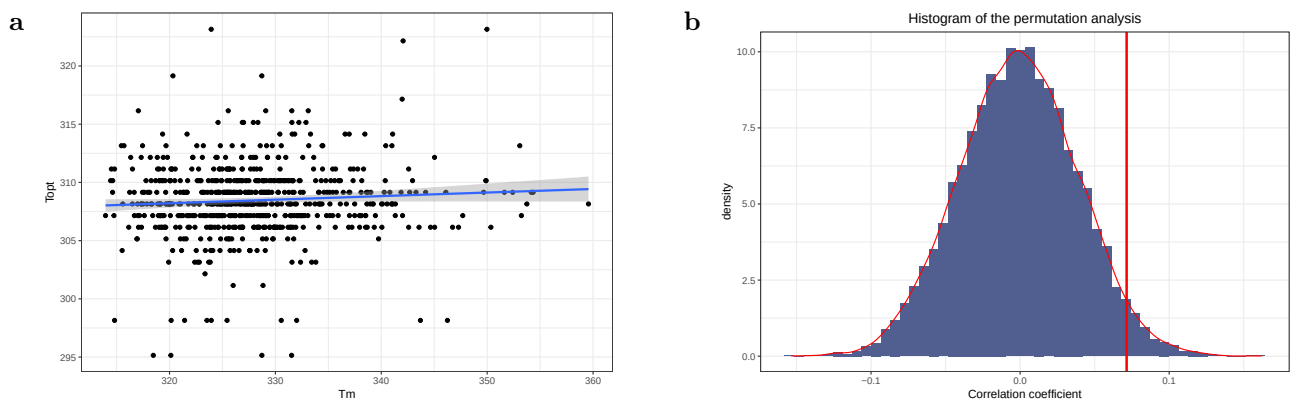


Figure S3: The correlation between initial parameter values of T_m and T_{opt}